

# BUSS386 Problem Set 4 — Solutions

## Pricing Forwards and Futures

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### Problem 1 — Cost-of-carry, non-dividend stock

(a)  $F_0 = 80 e^{0.04} = \boxed{\$83.26}$ .

(b) Market  $\$85 > \$83.26 \Rightarrow$  forward overpriced. Strategy: short forward, buy stock, borrow  $\$80$ .

| Action                     | CF at $t = 0$ | CF at $t = 1$     |
|----------------------------|---------------|-------------------|
| Buy stock at $\$80$        | $-80$         | $S_T$             |
| Short forward at $\$85$    | $0$           | $85 - S_T$        |
| Borrow $\$80$ at $r = 4\%$ | $+80$         | $-83.26$          |
| Net                        | $0$           | $\boxed{+\$1.74}$ |

(c) Market  $\$82 < \$83.26 \Rightarrow$  forward underpriced. Strategy: long forward, short stock, lend proceeds.

| Action                   | CF at $t = 0$ | CF at $t = 1$     |
|--------------------------|---------------|-------------------|
| Short stock (sell now)   | $+80$         | $-S_T$            |
| Long forward at $\$82$   | $0$           | $S_T - 82$        |
| Lend $\$80$ at $r = 4\%$ | $-80$         | $+83.26$          |
| Net                      | $0$           | $\boxed{+\$1.26}$ |

In both cases the profit is just the mispricing itself:  $|F^{\text{mkt}} - F_0|$ .

### Problem 2 — Discrete dividends + live valuation

(a) PV of dividends today:

$$I = 1 e^{-0.06 \cdot 2/12} + 1 e^{-0.06 \cdot 5/12} = 0.99005 + 0.97531 = 1.9654.$$

$$F_0 = (50 - 1.9654) e^{0.06 \cdot 0.5} = 48.0346 \cdot 1.03045 = \boxed{\$49.50}.$$

(b) At  $t = 3$  months: spot  $\$48$ , 3-month maturity remains, one  $\$1$  dividend in 2 months.

$$I' = 1 e^{-0.06 \cdot 2/12} = 0.9900.$$

$$F_t = (48 - 0.9900) e^{0.06 \cdot 0.25} = 47.0099 \cdot 1.01511 = \boxed{\$47.72}.$$

$$f = (F_t - F_0) e^{-r(T-t)} = (47.72 - 49.50) e^{-0.06 \cdot 0.25} = -1.78 \cdot 0.9851 = \boxed{-\$1.75}.$$

- (c) The contract was originally a fair (zero-value) trade, but as the stock fell from \$50 to \$48 the new market forward fell below the locked  $F_0 = \$49.50$ , so the obligation to buy at \$49.50 is now worse than the prevailing rate  $\Rightarrow$  negative value to the long.

### Problem 3 — KOSPI 200 index futures

(a)  $F_0 = 1,000 e^{(0.025-0.015) \cdot 4/12} = 1,000 e^{0.003\bar{3}} = \boxed{1,003.34}$ .

- (b) Market 1,010 > fair 1,003.34  $\Rightarrow$  futures overpriced. Strategy (per index unit):

| Action                                              | CF today | CF at $T = 4m$                                   |
|-----------------------------------------------------|----------|--------------------------------------------------|
| Buy $e^{-q \cdot 4/12}$ basket: $-1,000 e^{-0.005}$ | -995.01  | 1 unit basket worth $S_T$                        |
| Short futures at 1,010                              | 0        | $1,010 - S_T$                                    |
| Borrow 995.01 at $r = 2.5\%$                        | +995.01  | $-995.01 \cdot e^{0.025 \cdot 4/12} = -1,003.34$ |
| Net                                                 | 0        | $\boxed{+6.66 \text{ index pts}}$                |

Buying  $e^{-qT}$  baskets and reinvesting dividends gives exactly 1 basket at  $T$  (continuous-dividend trick).

- (c) Holding the index pays a continuous dividend stream at rate  $q$  that partially offsets the cost of borrowing  $S_0$  to buy spot — so the net carry is  $r - q$ , not  $r$ .

### Problem 4 — USD/KRW forward

(a)  $F_0 = 1,380 e^{(0.025-0.040) \cdot 0.5} = 1,380 e^{-0.0075} = \boxed{1,369.7 \text{ per USD}}$ .

- (b) USD is at a forward *discount* ( $F_0 < S_0$ ). KRW pays less interest than USD, so for there to be no arbitrage between (i) holding USD at  $r_f$  and (ii) holding KRW at  $r$  then converting at the forward, the forward FX rate must reduce the KRW value of USD.
- (c) Market 1,375 > fair 1,369.7  $\Rightarrow$  forward overpriced. Sell USD forward at the high market price, buy USD spot, finance with KRW borrow:

| Action (Korean investor)                              | CF today (KRW) | CF at $T = 6m$ (KRW)                              |
|-------------------------------------------------------|----------------|---------------------------------------------------|
| Buy $e^{-r_f \cdot 0.5}$ USD spot: $-1,380 e^{-0.02}$ | -1,352.67      | USD deposit grows to 1 USD $\rightarrow S_T$      |
| Short USD/KRW forward (1 USD at ₩1,375)               | 0              | $1,375 - S_T$                                     |
| Borrow ₩1,352.67 at $r = 2.5\%$                       | +1,352.67      | $-1,352.67 \cdot e^{0.025 \cdot 0.5} = -1,369.69$ |
| Net                                                   | 0              | $\boxed{+5.31}$                                   |

As a check, the profit is exactly the mispricing:  $1,375 - 1,369.69 = 5.31$ .

- (d) New rates  $(r, r_f) = (3.5\%, 4\%)$ :

$$F'_0 = 1,380 e^{(0.035-0.040) \cdot 0.5} = 1,380 e^{-0.0025} = \boxed{1,376.6 \text{ per USD}}.$$

Forward rises from ₩1,369.7 to ₩1,376.6 — the USD discount *narrows*. Higher KRW rate makes holding KRW relatively more attractive, so the USD does not need to be at as steep a forward discount to prevent arbitrage.

- (e) **Covered interest parity:** the forward FX premium (or discount) exactly offsets the interest-rate differential between two currencies, so it is impossible to earn a risk-free arbitrage by combining spot FX, the two money markets, and a forward.

## Problem 5 — Valuing a live forward

(a)  $F_0 = 40 e^{0.05} = 40 \cdot 1.05127 = \boxed{\$42.05}$ .

- (b) Three months in, 9 months remain.

$$F_t = 43 e^{0.05 \cdot 9/12} = 43 e^{0.0375} = 43 \cdot 1.03821 = \boxed{\$44.64}.$$

$$f_{\text{long}} = (F_t - F_0) e^{-r(T-t)} = (44.6431 - 42.0508) e^{-0.0375} = \boxed{\$2.50}.$$

- (c) Cross-check via the  $S_t$  formula, where the two discount factors collapse to  $e^{0.05 \cdot 3/12}$ :

$$f_{\text{long}} = S_t - F_0 e^{-r(T-t)} = 43 - 40 e^{0.0125} = 43 - 40.50 = \boxed{\$2.50}. \checkmark$$

- (d) With  $S_t = \$38$ :  $F_t = 38 e^{0.0375} = \$39.45$ , and

$$f_{\text{long}} = 38 - 40 e^{0.0125} = 38 - 40.50 = \boxed{-\$2.50}.$$

It *is* a mirror image, but not for the obvious reason. The long is worth exactly zero when  $S_t = F_0 e^{-r(T-t)} = \$40.50$  — that is the break-even spot, and value is simply the distance from it. \$43 sits \$2.50 above \$40.50; \$38 sits \$2.50 below. Note the two spot prices are *not* equidistant from the original \$40 (they are \$3 above and \$2 below): it is the break-even level, not the starting price, that they straddle.